12
Chapter

PSYCH 301
Cognitive Processes

Lecture 14:
Problem-Solving and Reasoning

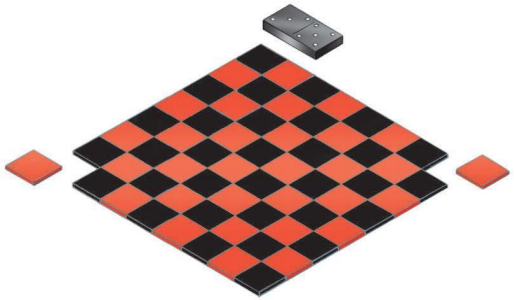
Fall 2026

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12 Chapter

Problem solving

- Problem solving as reorganization and insight
- Problem solving as search: a search that occurs between the posing of the problem and its solution
- There is more to problem solving than specifying the problem state
- How a problem is stated can affect its difficulty

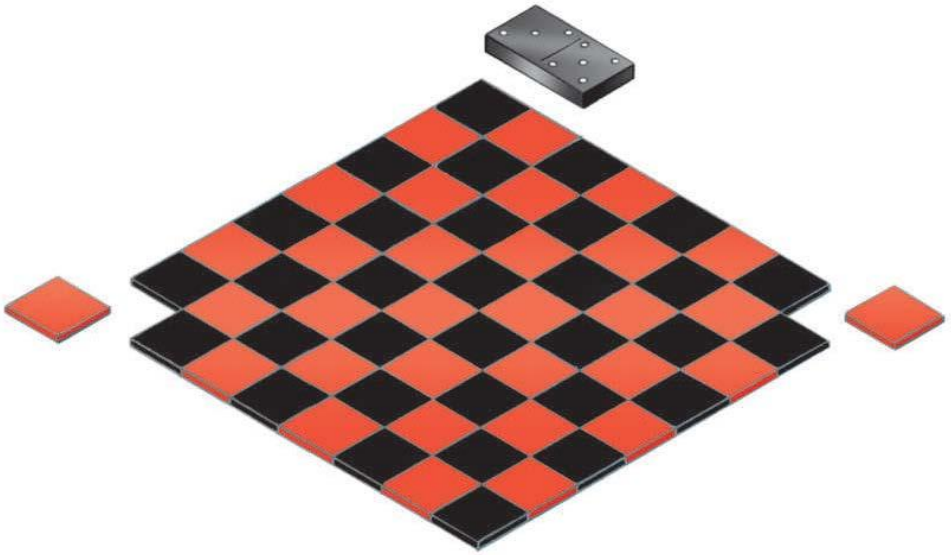


Using Analogies to Solve a Problem

- Using a solution to a similar problem guides solution to new problem
  - In a small Russian village, there were 32 bachelors and 32 unmarried women. Through tireless efforts, the village matchmaker succeeded in arranging 32 highly satisfactory marriages. The village was proud and happy. Then one drunken night, two bachelors, in a test of strength, stuffed each other with pierogies and died. Can the matchmaker, through some quick arrangements, come up with 31 heterosexual marriages among the 62 survivors?

12 Chapter

Using Analogies to Solve a Problem



Using Analogies to Solve a Problem

- Analogical problem solving: Using a solution to a similar problem guides solution to new problem
  - Russian marriage problem (source problem) → mutilated-checkerboard problem (target problem)
  - Analogical transfer: The transfer from one problem to another
    - Source problem to target problem
      - Target problem: the problem that S is trying to solve
      - Source problem: another problem that shares some similarities with the target problem
    - Learning about the Russian marriage problem enhances the ability to solve the mutilated-checkerboard problem

Using Analogies to Solve a Problem

- Duncker's radiation problem:
  - Try solving the following problem:
  - Suppose you are a doctor faced with a patient who has a malignant tumor in his stomach. It is impossible to operate on the patient, but unless the tumor is destroyed the patient will die. There is a kind of ray that can be used to destroy the tumor. If the rays reach it all at once at a sufficiently high intensity, the tumor will be destroyed. Unfortunately, at this intensity the healthy tissue the rays pass through on the way to the tumor will also be destroyed. At lower intensities the rays are harmless to healthy tissue, but they will not affect the tumor, either. What type of procedure might be used to destroy the tumor with the rays and at the same time avoid destroying the healthy tissue?

- Only 10% of S arrive at the correct solution

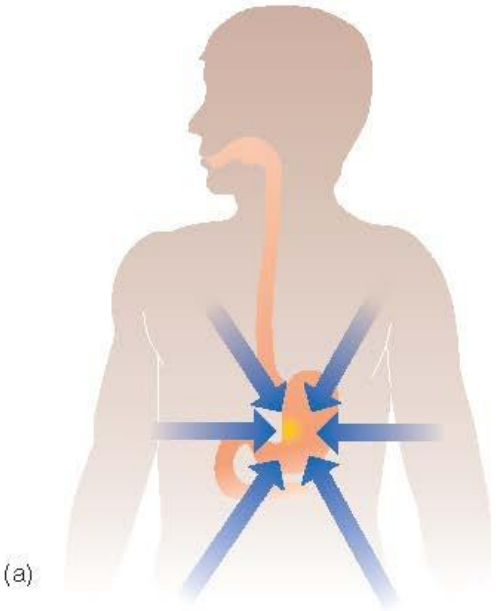
Using Analogies to Solve a Problem

- A small country was ruled from a strong fortress by a dictator. The fortress was situated in the middle of the country, surrounded by farms and villages. Many roads led to the fortress through the countryside. A rebel general vowed to capture the fortress. The general knew that an attack by his entire army would capture the fortress. He gathered his army at the head of one of the roads, ready to launch a full-scale direct attack. However, the general then learned that the dictator had planted mines on each of the roads. The mines were set so that small bodies of men could pass over them safely, since the dictator needed to move his troops and workers to and from the fortress. However, any large force would detonate the mines. Not only would this blow up the road, but it would also destroy many neighboring villages. It therefore seemed impossible to capture the fortress.

Using Analogies to Solve a Problem



(c)



(a)

Using Analogies to Solve a Problem

- After reading the fortress story, 30% of S arrive at the correct solution

- Gick and Holyoak
  - Noticing an analogous relationship
    - Crucial, and the most difficult step
  - Mapping correspondence between source and target
    - Elements of the source need to be mapped to element of the target
  - Applying mapping
    - to generate parallel solutions in target

- 70% of S where not able to see the correspondences
  - When S were told to think about the story they had read, success rate got to 75%

Using Analogies to Solve a Problem

- There are factors that facilitate the noticing and mapping steps
- One reason noticing is difficult is that people often focus on surface features
  - Fortress and tumor
- Can making surface features similar facilitate noticing analogous relationships?

Using Analogies to Solve a Problem

- In a physics lab at a major university, a very expensive lightbulb, which would emit precisely controlled quantities of light, was being used in some experiments. One morning Ruth, the research assistant, came into the lab and found that lightbulb no longer worked. She noticed that the filament inside the bulb had broken into two parts. The surrounding glass bulb was completely sealed, so there was no way to open it. Ruth knew that the lightbuild could be repaired if a brief, high intensity laser beam could be used to fuse the two parts of the filament into one.
- However, a high-intensity laser beam would also break the fragile glass surrounding the filament. At lower intensities the laser would not break the glass, but neither would it fuse the filament. What type of procedure might be used to fuse the filament with the laser and at the same time avoid breaking the glass?

Using Analogies to Solve a Problem

- Duncker's Radiation Problem used as the source
  - 81% of S that knew about the radiation problem solved the lightbulb problem
  - 10% of S in the control group were able to solve the problem

- Surface similarity helped in solving the problem

Using Analogies to Solve a Problem

- Lightbulb problem
  - High surface similarities aid analogical problem solving
    - Surface features: Specific elements of a given problem
  - What about structural features?
    - Underlying principle that governs the solution
      - Radiation: strong ray destroys tissue
      - Lightbulb: strong laser breaks lightbulb
    - Made two versions, keeping the surface features the same and varying the structural features

Using Analogies to Solve a Problem

- Source problem 1: Fragile-Glass version (source and target have similar structural features)
- *Problem*: A high-intensity laser beam would break the fragile glass surrounding the filament. At lower intensities the laser would not break the glass, but neither would it fuse the filament.
- *Ruth's Solution*: Ruth placed several lasers in a circle around the lightbulb and administered low-intensity laser beams from several directions at once. The beams all converged on the filament, where their combined effect was enough to fuse it. Because each spot on the surrounding glass received only a low-intensity beak from each laser, the glass was left intact.

- 69% of S who read this solution were able to solve the radiation problem

Using Analogies to Solve a Problem

- Source problem 2: Insufficient-Intensity version (source and target have different structural features)
- *Problem*: The laser generated only low-intensity beams that were not strong enough to fuse the filament. A much more intense laser beam was needed.
- *Ruth's Solution*: Ruth placed several lasers in a circle around the lightbulb and administered low-intensity laser beams from several directions at once. The beam all converged on the filament, where their combined effect was enough to fuse it.

- 33% of S who read this solution were able to solve the radiation problem

Using Analogies to Solve a Problem

- Making structural features more obvious aids analogical problem-solving
    - Structural features: The underlying principle(s) that govern the solution to a problem

- Analogical encoding: the process by which two problems are compared and similarities between them are determined
  - Effective way to get participants to pay attention to structural features that aide problem-solving
  - Having people compare source stories may be an effective way to force S focus on structural features

Using Analogies to Solve a Problem

- Analogical paradox
  - It can be difficult to apply analogies in the laboratory, but people routinely use analogies in real-world settings
  - Scientists use about 3 to 15 analogies in a 1-hour lab meeting

- In vivo problem-solving research
  - People are observed to determine how they solve problems in the real world
    - Advantage: naturalistic setting
    - Disadvantages: time-consuming, cannot isolate and control variables

How Experts Solve Problems

- What is an expert?
  - "A person who, by devoting a large amount of time to learning about a field and practicing and applying that learning, have become acknowledged as being extremely knowledgeable or skilled in that field."
    - Chess expert: someone who has spent 10,000 to 20,000 hours studying and playing chess
- Experts solve problems in their field faster and with a higher success rate than beginners
- But why are they better?

How Experts Solve Problems

- Experts possess more knowledge about their fields
- Chase & Simon (1973): compared how well a chess master and a beginner (fewer than 100 hours) were able to reproduce the positions of pieces on a chessboard after looking at an arrangement for 5 seconds
- Two conditions:
  - Pieces arranged in actual game positions
  - Pieces arranged randomly
- Experts did much better than non-experts in the first condition
  - They had stored these actual position in their LTM
  - The positions were accessed in chunks
  - These chunks were not available in the random condition

How Experts Solve Problems

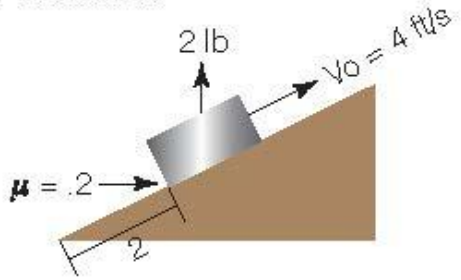
- Knowledge is organized so it can be accessed when needed to work on a problem
- Chi et al. (1982): presented 24 physics problems to a group of experts (physics professors) and a group of novices (students with one semester of physics)
  - They were asked to sort the problems into groups based on their similarities
  - Novices: sorted based on surface features (what the object looked like)
  - Experts: sorted based on structural features (the underlying principles involved)

How Experts Solve Problems

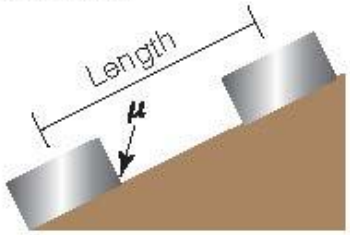
Novice

The novice grouped problems 23 and 24 together because they both involve similar objects (inclined planes).

Problem 23



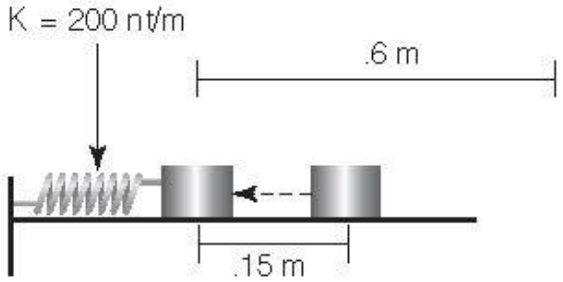
Problem 24



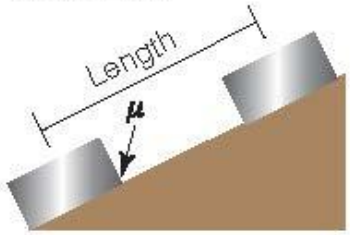
Expert

The expert grouped problems 21 and 24 together because they both involve similar physics principles (conservation of energy).

Problem 21



Problem 24



How Experts Solve Problems

- Experts spend more time analyzing problem
  - Slower start, but this strategy usually pays off
- Experts are no better than novices when given problems outside of their field

- Being an expert is not always an advantage
- Experts less likely to be open to new ways of looking at problems
  - Being an expert may be a disadvantage when confronting a problem that requires flexible thinking
    - Problems whose solution may involve rejecting the usual procedures in favor of other procedures that are less frequently used

Creative Problem-Solving

- Creativity
  - Innovative thinking
  - Novel ideas
  - New connections between existing ideas
  - Divergent thinking: open-ended; large number of potential "solutions"
- Problem solving as a process: most creative problem solving includes far more than just getting an idea
  - It involves a lengthy period of trial-and-error
  - Stages:
    - Problem generation: Problem finding, fact finding
    - Problem formulation: Problem definition, idea finding
    - Problem solving: Evaluation, planning
    - Solution implementation: taking action

Creative Problem-Solving

- Generating ideas is only one part of the creative process
  - Without ideas, the process stops
- Too much knowledge can sometimes hinder creative problem solving
  - Often problems require thinking flexibility and rejecting accepted procedures